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Knowledge contribution in virtual communities: accounting for multiple dimensions of social presence through social identity

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Integrating social presence theory and social identity theory, this study brings system design and social influence aspects together to explain their joint effects on knowledge contribution in virtual communities (VCs). Different from most prior information systems (IS) research that adopts a uni-dimensional approach and restricts social presence to be the subjective nature of media, we developed and empirically tested a model explaining the effects of multi-dimensional social presence on social identification processes and knowledge contribution. An online survey was conducted with four different VCs of interest. The results showed the difference in relative contribution of social presence dimensions on social identity as well as knowledge contribution. Both practical and theoretical implications are discussed.

Keywords: social presence; social identity; knowledge contribution; virtual communities

1. Introduction

Virtual communities (VCs) describe social aggregations with common values and interests on the Internet. This study is particularly interested in one type of VC, which is text-based online communities (like those listed in big-boards.com), which are also called online forums, message boards or discussion boards, (electronic) discussion groups, discussion forums, bulletin boards and Internet forums. Such VCs have been widely adopted to support knowledge generation and knowledge sharing, bringing great commercial and social value. Similar to their physical counterparts, members' knowledge contribution is the pulse for generating content and building relationships (Lee *et al*. 2003). In VCs, such contribution is usually indicated by the number of postings or/and number of created threads (Miranda and Saunders 2003). A recent study indicates that active on-going contribution is ranked as one of the top factors for a successful VC (Leimeister *et al*. 2004). How to support social interaction and encourage knowledge contribution in VCs becomes a major challenge for practitioners and attracts much academic interest as well.

Identification, an important identity process reflecting influences exerted from the collective, has been demonstrated to be a critical factor in predicting prosocial behaviour in general, and community participation in particular (Bagozzi and Dholakia 2002). Identification means that the individual defines him/herself in terms of the membership in the group. The resulting perceptions of oneness with or belongingness to the group provide a more autonomous motivation resulting not only in a higher quality of engagement (e.g. greater persistence, effort, etc.) but also in more positive experiences such as enjoyment, sense of purpose and well-being (Ryan and Deci 2001). Numerous empirical evidence has shown that identification enhances cooperative behaviour (Dukerich *et al*. 2002), commitment (Foreman and Whetten 2002), VC participation (Dholakia *et al*. 2004), knowledge contribution (Kankanhalli *et al*. 2005), customer extra role behaviour (Ahearne *et al*. 2005) and organisational citizenship behaviour (Van Der Vegt and Bunderson, 2005). Moreover, adoption of certain social exchange forms and social norms is contingent upon the individual's identification with the group (Constant *et al*. 1994, Flynn 2005).

Despite the importance of identification, most prior studies on VCs only incorporate identification or social identity as an antecedent or moderator to explain members' behaviour (Constant *et al*. 1994, Kankanhalli *et al*. 2005). Some other studies with a focus on 'community' directly apply the existing theories developed in the physical context to VCs, e.g. social influence theory (Bagozzi and Dholakia 2002), where the impacts of IT have been under-studied. For VCs, the technological capacity of communication media plays a critical role in determining individual behaviour, as all the psychological experience of the

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represented places and people is computer-mediated (Jones *et al*. 2004). We still lack the understanding about how identification with a VC develops in general, and the impacts of IT artefacts in particular.

In this article, we attempt to bridge the gap in prior research by exploring the system design effect on identification and how system design and identification jointly influence knowledge contribution in VCs. On the system design side, social presence emerges as a major design principle in telecommunication systems as well as a core construct in studying computer-mediated communication (Biocca *et al*. 1995). First proposed by Short *et al*. (1976), social presence refers to the 'subjective qualities' of the medium and was conceptualised as a uni-dimensional construct. This conceptualisation has been widely employed in IS research (Kumar and Benbasat 2002, Miranda and Saunders 2003). However, recent developments in the present research have provided rich evidence suggesting the multi-dimensional nature of social presence (Lombard and Ditton 1997). Social presence, the social aspect of presence, is re-conceptualised as the fluctuating phenomenal nature of the medium; that is, the properties of the communication interaction specifically rather than direct attributions about the medium *per se* (Biocca *et al*. 2003). This advancement suggests the significant impact of developers in not only the interface design but also the experience design (Pares and Pares 2006). Although insightful, the multi-dimensional conceptualisation of presence has been mainly limited to the investigation of immersive media, e.g. virtual reality. Shen and Khalifa (2007) examined the appropriateness of multi-dimensional conceptualisations of social presence for studying VCs, a different media featured with non-immersive technologies, and proposed a multi-conceptualisation of social presence for the VC context consisting of three dimensions; awareness, affective social presence and cognitive social presence.

Integrating social identity theory and social presence theory, we developed and empirically tested a model explaining the effects of multi-dimensional social presence on the social identification process and knowledge contribution in VCs. A rigorous mediation test with four VCs was used to empirically validate the multi-dimensional conceptualisation of social presence and elucidate the relationship between social presence and social identity in affecting knowledge contribution. This study advances the research on social presence by adopting a multi-dimensional approach and providing the empirical evidence for its validity and utility. Our results also enhance the understanding of the distinct role of social presence in social identity formation and knowledge contribution in VCs. Finally, this research contributes to social identity literature by demonstrating the antecedents for social identity in the VC context. Practically, the findings are expected to provide design guidelines for VC designers and administrators so as to motivate the community members' contribution.

This article is structured as follows. We first discuss social identity and self-categorisation theories with respect to VC participation. This is followed by the description of the conceptualisation of social presence proposed by Shen and Khalifa (2007). Following is a theoretical development section where the research model is introduced and hypotheses are justified. Then we describe the empirical study and discuss the results. Finally, we summarise the study, discuss the theoretical and practical implications and make suggestions for future research.

2. Social identity and self-categorisation theory

Much evidence has shown that the maximisation of self-interest does not adequately explain why people contribute to VCs as opposed to lurking or free-riding (Wasko and Faraj 2000). As membership and the extent of participation in VCs are totally volitional and may be terminated by the member relatively effortlessly (Bagozzi and Dholakia 2002), the main motivation for contributing to a VC relies on members' passion, and commitment and identification (Wenger and Snyder 2000). Also, the adoption of certain social exchange norms, e.g. reciprocity, is contingent upon the individual's understanding of the context (Constant *et al*. 1994), which is mainly reflected through members' identities (Weber *et al*. 2004). Thus, the social identity theory and self-categorisation theory (Oakes 1987, Ashforth and Mael 1989) have been advocated in explaining members' contribution (Bagozzi and Dholakia 2002). These theories assume that the group can obtain cohesiveness through encouraging the individual to define him/herself in terms of membership in the group. Social identity refers to one's conception of self in terms of the defining features of a self-inclusive social category that renders the self stereotypically 'interchangeable' with other group members, and stereotypically distinct from outsiders (Hogg and Abrams 1988). The resulting perceptions of oneness with or belongingness of the group provide a more autonomous motivation that brings not only a higher quality of engagement (e.g. greater persistence, effort, etc.) but also more positive experiences such as enjoyment, sense of purpose and well-being (Ryan and Deci 2001). Particularly, several studies have demonstrated the important role of social identity in motivating members' contribution in VCs (Bagozzi and Dholakia 2002, 2006, Kankanhalli *et al*. 2005).

However, most of the prior studies take social identity as given without addressing its antecedents.

We still lack knowledge regarding how to enhance members' identification. According to self-categorisation theory, one crucial determinant of social identity is the fit between a social categorisation and reality in both its comparative and normative aspects (Forehand *et al.* 2002). Comparative fit means that a set of stimuli are more likely to be categorised as a single entity when the intraclass differences between those items are seen to be smaller than the interclass differences between those items and others that are included in a given comparative context. Normative fit refers to the content-related aspects of the match between category specification and the instances being represented. In addition, both types of fit determine social identification in interaction with perceiver readiness or cognitive accessibility (Haslam *et al.* 1999). Accessibility is the readiness of a given category to become activated in the person. It is a function of the person's current tasks and goals and of the likelihood that certain objects or events will occur in the situation (Oakes *et al.* 1994). This means that people are more likely to define themselves in terms of a particular identity to the extent that it has prior meaning for them and it matches subjectively relevant comparative and normative features of reality.

Of more particular interest to this study is that these theories suggest that contextual stimuli play an important role in making certain aspects of the self salient, leading to a dynamic and fluctuating self-definition. In the context of VCs, the context stimuli are either from the technical context of the VC or from the other members socialising in the VC. All stimuli are presented through the interface and subject to the perception and reflection of each individual. The resulting virtual experiences that an individual develops will provide the information regarding comparative and/or normative fit.

3. Conceptualisation of social presence

Social presence refers to the feeling of 'being with another' or the degree of salience of the other person in the interaction and the consequent salience of the interpersonal relationships (Biocca *et al.* 2003). It is considered as the central design principle for social computing technologies, e.g. Multi-User Dungeons (MUDs), email, online chat and online communities (Ijsselsteijn and Riva 2003). Even for some new web-based applications, social presence remains critical in understanding user behaviour. For instance, several studies reported that social presence plays a role in affecting tagging behaviour with a bookmark managing system (Lee 2006), mobile and online media (Ames and Naaman 2007), Flickr (Nov *et al.* 2008) and in Facebook usage (Chui *et al.* 2008). Social presence has been demonstrated to underpin the key psychological processes, e.g. social orientation (Lee 2006), public, family and friends' motivations (Nov *et al.* 2008) and we-intention (i.e. group intentions in the form 'I intend we act') (Chui *et al.*, 2008). In most prior research, a uni-dimensional conceptualisation of social presence (Short *et al.* 1976) is adopted, and the social presence is considered as stable properties of the medium (see Biocca *et al.* (2003), for a review).

However, this has been challenged by many present researchers who argue for a multi-dimensional approach in understanding the social presence. It is widely acknowledged that presence should be considered as an experience that varies in a moment-to-moment fashion (Heeter 2003). Biocca *et al.* (2001) even re-conceptualised social presence as the fluctuating phenomenal nature of the medium, that is properties of the communication interaction specifically rather than direct attributions about the medium per se. From a design point of view, Pares and Pares (2006) also argued that a successful virtual environment will not only provide the indistinguishable sensory experiences, but will also need to mimic the human perception and reception. Instead of focusing on artefact design, they advocate the experience design. A multi-dimensional approach is advocated to better capture the user experience and accordingly provides more integrated practical implications for designing the virtual experience.

On the basis of a comprehensive review of social presence literature and the related studies, Shen and Khalifa (2007) proposed a three-dimensional conceptualisation of social presence for VCs. They defined social presence as the awareness of the other sentient beings accompanied by affective and cognitive engagement with others in computer-mediated social spaces. Three dimensions (i.e. awareness, affective social presence and cognitive social presence) are identified to capture a user's holistic experience with the computer-mediated social space. Awareness refers to the extent to which a user believes other social actors appear to exist and are able to react to the user (Heeter 1992). In online communities, users communicate with the keyboard. The sensory extensity and intensity are both low. Showing the user status (e.g. online/offline, where is he/she, or what is he/she doing) and using self-presentation features (e.g. images and avatars) to enhance the awareness. In addition, awareness in an online community is also achieved through users' continuous participation in online discussion in the form of postings. Affective social presence refers to the extent to which a user's emotional connection is aroused by virtual social interaction with others. Huang *et al.* (1999) argue that people do not think about being present in the real world – they feel that they are. Emotions, at the very least, are a prerequisite to experience being present in a virtual environment.

Vastfjall (2003) also found that the sense of presence and emotional reactions were highly interrelated and speculate that the subjective sense of presence is not a separate construct from emotional reaction, but a feeling of presence is actually an emotion. An emotional reaction may be used as evidence of a participant's social presence in an environment and as information input for further evaluation and behavioural response. Cognitive social presence refers to the extent to which a user is able to construct and confirm meaning about his/her relationship with others and the social space. The presence theorists have employed theories of cognition and memory to understand the nature of the cognitive processes that lead to the sense of presence. In general, the user has to be aware of the meaning of the VC in order to be 'present' in the context offered by a symbolic system (Riva *et al.* 2002).

4. Research model and hypotheses

Social influences have been shown to be critical to prompt individuals' knowledge contribution in VCs (Wasko and Faraj 2005). Perceptions of shared social identity provide people with multiple motivations for knowledge sharing and also with a shared cognitive framework that allows the sharing behaviour to be mutually beneficial and productive (Postmes 2003). The development of social identity with a VC is dependent upon the members' virtual experiences that provide the information regarding comparative and/or normative fit. Adopting a multiple dimensional approach, social presence dimensions capture the member's experience with VCs. Thus, relying on the social identity and social presence theories, we developed the research model as shown in Figure 1. The justification for the hypotheses will be discussed as follows.

4.1. Social identity and knowledge contribution

According to social identity theory, social identification leads individuals to perceive themselves in terms of the characteristics they share with other members rather than their distinctive characteristics (Turner 1982). It turns the group, psychologically, into a part of the self, blurring the distinction between the self and the group (Smith and Henry 1996). In this study, the social identity with a VC is defined as the emotional significance that the members attach to their membership in that community (Van Der Vegt and Bunderson 2005). Such an emotional social identity, also characterised as affective commitment (Allen and Meyer 1996), has been shown to most clearly 'supply the motivational force' leading to action or the 'readiness to engage in or disengage from interaction' (Bergami and Bagozzi 2000, p. 563).

Nahapiet and Ghoshal (1998) argue that social identity nurtures one's motivation to share knowledge, in contrast, distinct and contradictory identities within communities set up barriers to knowledge sharing. Social identity, implying an emotional involvement with VCs, fosters loyalty and citizenship behaviours (Ellemers *et al.* 1999, Bergami and Bagozzi 2000, Bagozzi and Dholakia 2002). Such identification renders individuals to maintain a positive self-defining relationship with other members, and elevate their activeness to knowledge contribution (Hogg and Abrams 1988). Bagozzi and Dholakia (2002) further confirm that social identity motivates the participation in online interaction by enhancing 'we-intention', i.e. individual commitment to participate in a joint action, and involves an implicit or explicit agreement between the participants to engage in that joint action. Chiu *et al.*' s (2006) empirical study also demonstrates that social identity is able to increase the volume of knowledge sharing in VCs. Consistent with the prior research, we hypothesise that:

> **H1:** A member's social identity with a VC will be positively related with his/her knowledge contribution to the VC.

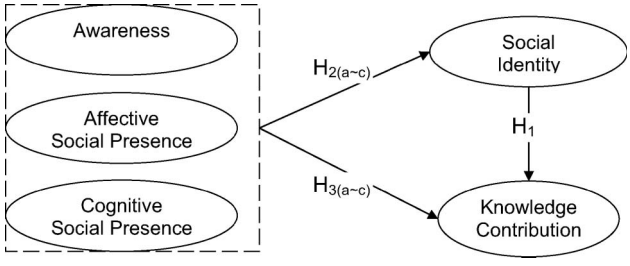

Figure 1. Research model.

4.2. Social presence and social identity

According to the social identity theory, one determinant for developing social identity is the comparative fit which means a set of stimuli are more likely to be categorised as a single entity when the intraclass differences between those items are seen to be smaller than the interclass differences between those items and others that are included in a given comparative context. The awareness of the others' existence and the social context has been demonstrated as a prerequisite for conducting social activities and maintaining a social community (Prasolova-Forland 2002). A user with a stronger sense of awareness of the other members as well as the community will be more likely to develop comparative fit than one with low awareness. Therefore, we hypothesise that:

H2a: A member's sense of awareness will be positively related to his/her social identity with the VC.

Identification with a social group is mainly derived from the group's ability to fulfil its members' needs (Hogg and Abrams 1988). The sense of affective social presence refers to the users' emotional connection with others and the online community. This dimension is closely tied with the feeling of intimacy and immediacy with others (Short *et al*. 1976). Reis and Shaver (1988) found that the feeling of intimacy was positively related to the tendency of self-disclosure, perception of partner disclosure and of partner responsiveness. The sense of immediacy or perceived distance among communicators contributes to a positive and affective view of the others (Christophel 1990, Richmond 1990). Thus, online community members with a salient sense of affective social presence will perceive a high likelihood of exchanging useful information and developing a social relationship. Therefore, we hypothesise that:

H2b: A member's sense of affective social presence will be positively related to his/her social identity with the VC.

Normative fit refers to the content-related aspects of the match between category specification and the instances being represented. It serves another important condition for social identity development (Forehand *et al*. 2002). Cognitive social presence means the perceived mutual understanding existing among communicators. It also suggests that users can make sense of the online community (e.g. purpose of the community). Users with a strong sense of cognitive social presence will be more likely to identify the common ground within the online community and to carry a more effective, efficient and satisfying communication with the others in the VC (Daft and Lengel 1986), leading to the high likelihood of developing social identity with the VC. Therefore, we hypothesise that:

H2c: A member's sense of cognitive social presence will be positively related to his/her social identity with the VC.

4.3. Social presence and knowledge contribution

In addition to indirect effects of social presence on knowledge contribution, we also propose a direct effect of social presence on members' contribution. Social identity theory interprets knowledge contribution as an intentional behaviour resulting from the identification process (Bagozzi and Dholakia 2002), which may not capture impulsive or automatic aspects of individual behaviour. As demonstrated in social psychological research, social stimuli can trigger the automatic, non-conscious perceptual interpretation that serves as the basis for further responses to the situation (Baumeister 2002). Bargh (1997) finds that 'there is insufficient evidence to support the position that conscious mediation of situational effects is the rule rather than the exception' (p. 5). Social presence serves as an interface between participants and VCs and reflects the physiological and psychological perception of the represented social environment. The stronger the sense of social presence, the more approximate is the VC to a real one. The salient social stimuli may activate the direct access processing of existing goals or trigger an automatic perceptual interpretation, both of which do not necessarily involve a well-developed intention. Consequently, we hypothesise that:

H3a: A member's sense of awareness will be positively related to his/her knowledge contribution to the VC.

H3b: A member's sense of affective social presence will be positively related to his/her knowledge contribution to the VC.

H3c: A member's sense of cognitive social presence will be positively related to his/her knowledge contribution to the VC.

5. Method

5.1. Sample and data collection

In order to test the research model, an online survey was conducted with four different VCs of interest. Because the characteristics of VCs such as nature (community of practice vs. community of interest), origin (network vs. small-group-based communities), size and life-cycle stage (forming, norming, storming, performing, adjourning) may potentially affect the results, they need to be controlled. The four selected VCs were similar in terms of (1) software package; (2) size; (3) community age; (4) content organisation and (5) theme. All of the VCs were using the same software package of vBulletin[1] with similar interface design and functions, e.g. browsing, searching, synchronous and asynchronous discussion, multi-media exchanging and voting. Members could attach animated icons (e.g. facial expressions) to better express their feelings, as well as audio and video to enrich their exchanges. They also shared a similar community design in terms of policies and organisation structures. All discussions were structured with various boards, e.g. general social discussion, sports, fashion and shopping. The selected communities had been operational for 2–3 years with accumulated registered membership ranging from 21,000 to 310,000. However, at the time we conducted this research, active members[2] were around 3000–5000.

Finally all selected VCs did not have any association with any physical social entities.

In these VCs, the knowledge in communities is mainly contributed by and only accessed by registered members. We obtained the permission of the management of the VCs to post a link to our online questionnaire for 3 days, promising to share the results of the survey with them. A total of 430 registered members provided valid responses. For each of the respondents, we retrieved from the database his/her knowledge contribution data for two weeks starting from the date that he/she submitted the survey. Table 1 shows the demographic statistics of the respondents and Table 2 reports their actual contribution information as retrieved from the database.

The possibility of non-response bias was examined by comparing our sample with a random sample of the total population in terms of tenure of membership, gender and age. There was no significant difference, indicating the representativeness of our sample. Group difference was tested with ANOVA for all constructs. No significant differences were found, which allowed us to pool together the data from different communities.

5.2. *Measures*

With a similar approach in extant studies (Miranda and Saunders 2003, Wasko and Faraj 2005, Chiu *et al.* 2006), we objectively examined the knowledge contribution in VCs based on the message postings: (1) the total number of postings, (2) the number of different threads where the postings were made and (3) the number of new threads created. The measurement test demonstrated that these three items were highly correlated and we treated them as reflective measures in this study. We measured awareness, affective and cognitive social presence with a seven-point scale adapted from Shen and Khalifa (2007). Two items for awareness captured how frequently the participant was aware of others as well as others being aware of him/her. Two measures for affective social presence focus on the individual's emotional involvement with others and perceived emotional connections in the online community. Three items for cognitive social presence captured the extent to which the users made sense of the social context. As for social identity, two measures were adapted from the organisational identification (OID) scale developed by Mael and Ashforth (1992). They are: 'When I talk about this forum, I usually say "we" rather than "they"'; and 'When someone praises this forum, it feels like a personal compliment'. Compared with the other multidimensional instruments, e.g. Cameron (2004), the OID instrument has been widely adopted in organisational research (van Knippenberg 2000, Dukerich *et al.* 2002) as well as information system research (Ma and Agarwal 2007).

5.3. *Data analysis*

Structural equation modelling was used to test the models shown in Figure 1. The software of AMOS 5.0 was employed for this purpose. We followed a two-step procedure based on the approach recommended by Anderson and Gerbing (1988). We first tested the measurement model with confirmatory factor analysis (CFA). When the measurement model held, a structural model was examined.

6. Results and discussion

Before checking the measurement and structural models, we relied on Harman's single-factor, a widely used method, to check for common method variance that may threaten the internal validity (Podsakoff 1986). According to this approach, common method

Table 2. Information of membership activities.

	Min.	Max.	Mean	Std.
Number of postings	0	1126	53.54	138.708
Number of threads initiated	0	205	2.86	13.042
Number of threads participated	0	687	36.44	86.927
Tenure of the membership (Day)	0.449	1168.648	360.483	257.594

Table 1. Demographic information.

Variables	Items	%	Variables	Items	%
Gender	Male	83.3	Age	<20	60.9
	Female	16.7		20–35	36.3
Level	Member	96.0		>36	2.8
	Manager	4.0	Frequency of login	Less than once a month	1.2
Familiarity with VC	Not at all	0.7		Once a week	5.1
	Moderate	32.6		Several times a week	16.7
	Fair	44.7		Once a day	23.5
	Very much	22.1		Several times a day	53.5

variance is present if a single factor accounts for the majority of the covariance in the dependent and independent variables. In this study, there was no dominant factor emerging from the factor analysis, implying that common method variance was not a serious problem.

6.1. Measurement model

We conducted CFA to assess the construct reliability and validity. The constructs' reliability was assessed with Cronbach's α and composite reliability (Fornell and Larcker 1981). The Cronbach's α values ranged from 0.726 to 0.880, and composite reliability from 0.600 to 0.915, indicating an adequate reliability (Hatcher 1994). Convergent validity was assessed by checking the loadings. Table 3 reports the measurement model results. As shown in Table 3, the estimated standard loadings ranged from 0.591 to 0.927, above the commonly used cut-off value of 0.50 (Teo and King 1996). All loadings were significant at $p \leq 0.001$ level, indicating good convergent validity.

Discriminant validity was assessed by comparing the square root of average variance extracted (AVE) for a particular construct to its correlations with the other constructs (Fornell and Larcker 1981). Table 4 shows the correlations between constructs and values on the diagonal were the square roots of the AVE scores which were all higher than the correlations among the constructs, demonstrating discriminant validity.

Noticing the high correlations among social identity, affective social presence and knowledge contribution, we conducted chi-square difference tests for each pair of constructs. The chi-square differences between the unconstrained and constrained model were all significant at the 99.9% level, demonstrating that each pair of constructs was much better when treated as distinct (Bagozzi and Phillips 1982, Anderson and Gerbing 1988). Furthermore, all items loaded higher on their respective constructs than others, providing additional support for discriminant validity. The discriminate analysis results empirically verified our multi-dimensional conceptualisation of social presence, implying distinct aspects of social presence experienced by VC members.

6.2. Structural model

Table 5 presents the results of structural models. The initial full model fits well overall as indicated by the goodness of fit indices, e.g. goodness-of-fit index (GFI), adjusted goodness-of-fit index (AGFI) and comparative-fit index (CFI), above the threshold values (Hair *et al.* 1998) and the root mean square error of approximation (RMSEA) less than 0.05 (Browne and Cudeck 1993).

To obtain further support for the validity of the model, rather than using a saturated model where 'everything is related to everything' as the baseline, we performed formal tests of mediation for social identity. Specifically, we used the Aroian version of the Sobel test suggested by Baron and Kenney (1986). The test

Table 3. Measurement loadings and reliability.

Constructs	Standard loading	Reliability
Knowledge contribution		0.915[a]
KC1	0.923*	
KC2	0.800*	
KC3	0.927*	
Social Identity		0.619[a]
SI1	0.744*	
SI2	0.591*	
Awareness		0.600[a]
Awareness1	0.684*	
Awareness2	0.615*	
Cognitive social presence		0.859[a]
CSP1	0.884*	
CSP2	0.824*	
CSP3	0.743*	
Affective social presence		0.833[a]
ASP1	0.812*	
ASP2	0.877*	

*$p <= 0.001$.
[a]Composite reliability.

Table 4. Correlations between latent constructs.

	KC	SI	CSP	ASP	A
Knowledge contribution (KC)	0.885				
Social identity (SI)	0.611	0.672			
Cognitive social presence (CSP)	0.419	0.367	0.819		
Affective social presence (ASP)	0.441	0.504	0.356	0.845	
Awareness (A)	0.198	0.051	0	0	0.650

The diagonal numbers are the square root of AVE (Fornell and Larcker 1981).

Table 5. Goodness of fit indices for the structural model.

Goodness of fit indices	Initial model	Revised model	Desired levels
χ^2 (*df*)	40.795	41.257	Smaller
df	45	46	–
χ^2/df	0.907	0.897	<3
p	0.650	0.671	–
GFI	0.985	0.984	>0.90
AGFI	0.973	0.974	>0.80
CFI	1	0.984	>0.90
RMSEA	0	0	<0.05

results confirmed that social identity mediated the effects of affective social presence ($t = 6.157$; $p < 0.001$) and cognitive social presence ($t = 5.743$; $p < 0.001$) on community members' knowledge contribution. However, the mediating effects of social identity on the relationship between awareness and knowledge contribution were non-significant ($t = -0.242$; $p = 0.809$). Figure 2 is the revised model without the link between awareness and social identity. The revised model performed as good as the initial one, except for slight difference in terms of GFI and CFI, but more parsimonious (see Table 5). The discussion will be based on the revised model.

The revised model explained 46.2% of the variance of knowledge contribution, and 29.7% of the variance of social identity, providing strong support for our research model. Consistent with prior research, social identity was found to be a strong driving force for members' knowledge contribution (H1: 0.468, $p < 0.01$). The affective social presence contributed more to social identity with much higher magnitude of effect (H2b: 0.432, $p < 0.01$) than that of cognitive social presence (H2c: 0.212, $p < 0.01$). Social identity implies members' emotional attachment to the communities, thus its formation is more likely to be triggered by the VC design that can arouse members' emotional involvement in VC communication. Our results also confirmed the significant relationship between the sense of cognitive social presence and social identity. A better mutual understanding among communicators catalyses the development of social identity. The relationship between awareness and social identity was found to be non-significant (H2a rejected). Awareness, shaping the community members' automatic perception of the social environment of a VC, is not sufficient to enlighten the members' social identification process, but still can directly activate the members' knowledge contribution behaviour. This confirms Bargh's (1997) finding that conscious mediation of situational effects is not a rule.

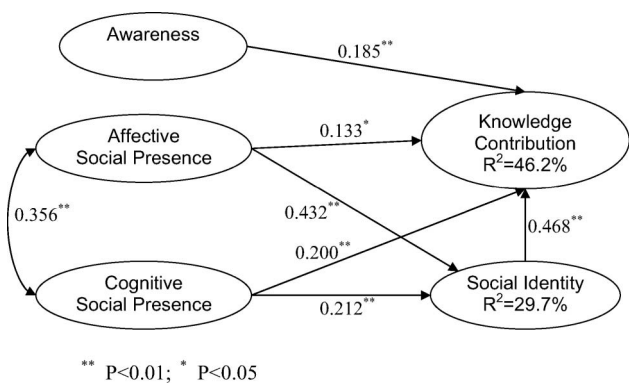

Figure 2. Structural results of the revised model.

The direct impacts of social presence on knowledge contribution varied for different dimensions. The awareness dimension and cognitive social presence had a higher magnitude of effect on members' contribution (H3a: 0.185, $p < 0.01$; H3c: 0.200, $p < 0.01$, respectively) than that of affective social presence (H3b: 0.133, $p < 0.05$). In the context of a VC, the sense of virtual environment and the shared understanding among community members may make them believe the ability of VC to fulfil their social needs, and therefore are more likely to elevate the knowledge contribution. In contrast, the effect of affective social presence on knowledge contribution is more likely mediated through social identity.

7. Implications and future research

In this study, we integrate social presence theory and social identity theory to explain their joint effects on knowledge contribution in VCs. Our study advances the research on social presence by adopting a multiple-dimensional approach and providing the empirical evidence for its validity and utility. The results also contribute to social identity literature by examining the effect of social presence on social identity in a VC context.

7.1. Theoretical implications

This research entails several theoretical implications. First, we investigated social presence as an important antecedent for social identification in VCs, contributing to both VC literature and identification formation research. Social identification has been studied in many contexts, e.g. groups, organisations, communities, and more particularly VCs. Extant evidence shows social identification as an important intervention for collective action (Turner *et al.* 1994). However, most prior studies take social identification as given without shedding light on how to enhance social identification among members. Even in physical contexts, social identification formation remains an under-studied area. To date, most research on this topic has been done in the context of formal organisations (Dukerich *et al.* 2002) and focuses on theoretical development, providing little empirical evidence (Foreman and Whetten 2002). In the context of physical communities, studies on social identification have just started and many controversies remain in several fundamental areas, such as the existence of community identities (Puddifoot 2003). In the IS field where VCs have received much interest, most prior studies take social identification or theories developed in organisational contexts as given without exploring the specificity brought by IT artefacts. This study advances the

theoretical work on social identification by investigating the role of social presence in shaping members' social identification.

Second, deviating from most prior IS research, we adopt a multi-dimensional conceptualisation of social presence and our results demonstrate a great potential for this conceptualisation in understanding VC phenomena. Although social presence has been widely adopted by most prior research to explain system design effects and user behaviour in online settings, e.g. e-commerce and online communities, social presence is limited to the 'subjective nature' of the medium and is conceptualised as unidimensional. Such a conceptualisation has been challenged by both presence and media studies as it has failed to capture the complexity of user experiences with the medium. It is argued that social presence should be conceptualised as the properties of virtual communication with the media. Shen and Khalifa (2007) tailored the multi-dimensional conceptualisations originally developed in immersive technical contexts to understand non-immersive VC contexts. They also demonstrated how different dimensions of social presence influence motivation and subsequent VC participation. Our results contribute to the research on this multi-dimensional conceptualisation of social presence by showing its effects on social identity. Particularly, only affective and cognitive social presence contributes to the development of social identity; while the mere sense of awareness is not sufficient to strengthen identification. This implies that virtual experiences captured with different dimensions of social presence may elicit different psychological and/or social psychological processes. Future research may examine the multiple channels to have a more holistic understanding about the effect of social presence on knowledge contribution in VCs.

7.2. *Practical implications*

In terms of practical implications, the multi-dimensional conceptualisation of social presence extends the design objectives for VC and provides the roadmap for engineering online users' experiences. In the past, most practices aimed at supporting social presence focused on awareness, e.g. the translucent system. The multiple components inherent in social presence suggest that system design needs to go beyond the basic awareness requirement to focus on affective and cognitive aspects. Furthermore, by adopting a multi-dimensional conceptualisation of social presence, VC designers and managers may obtain a better understanding of the driving force for community contribution and psychological mechanisms (e.g. social identification). Some VCs may be more affective (e.g. community for social support) while others may be more cognitive (e.g. communities of practice or open-source communities). A multi-dimensional conceptualisation of social presence will be able to discern such inherent differences and provide more specific design guidance. In this research, we demonstrate that the driving forces for social identity are mainly affective social presence. Future research should continue exploring the role of social presence in the other important social/psychological processes.

7.3. *Limitations and future research*

Certain limitations need to be noted in interpreting our results. Although the current findings are encouraging and supportive, the context of our study, i.e. communities of interest, may limit the generalisability of results. We controlled the characteristics of the VC such as their nature (community of practice vs. community of interest), origin (network vs. small-group-based communities), size and life-cycle stage (forming, norming, storming, performing, adjourning), which potentially affect the results. Future research could test the model in a variety of contexts. Moreover, with a cross-sectional design, we cannot empirically establish causality between social presence and social identity. A longitudinal approach will be very helpful in understanding the interaction between social presences, formation of social identity and knowledge contribution in VCs. Finally, we are aware of various multi-dimensional instruments of social identity, e.g. centrality; ingroup affect; and ingroup ties by Cameron (2004), cognitive, affective and evaluative social identity by Bergami and Bagozzi (2000). It will be interesting to examine how different dimensions of social presence affect different dimensions of social identity in the future research as well.

8. Conclusions

By integrating social presence theory and social identity theory, this study bridges the gap in prior research by exploring the system design effect on identification and how system design and identification jointly influence knowledge contribution in VCs. Different from most prior IS research that adopts a uni-dimensional approach and restricts social presence to be the subjective nature of media, we developed and empirically tested a model explaining the effects of multi-dimensional social presence on social identification processes and knowledge contribution. The results indicated different effects of social presence dimensions on social identity and knowledge contribution, and validated the mediating effects of social identity on the links between social presence dimensions and

knowledge contribution. This study suggests the necessity of employing multi-dimensional approach in social presence research and online community research in general.

Notes

1. Such online forums are primarily text-based and members can also use emoticons, avatar and formatting tools for interaction (see http://www.adminfusion.com/demos.php?action=view&id=6 for features supported by vBulletin).
2. According to the policies of the online forums, saying a member is active means this member logged in at least once within the last month. Despite the large amount of registered members, only those active ones constitute the current state of those online forums. We observed the number of users who were online at different times within a day for 1 week. Then the average was used as the estimation of the amount of active members for each online forum.

References

Ahearne, M., Bhattacharya, C.B., and Thomas, G., 2005. Antecedents and consequences of customer-company identification: expanding the role of relationship marketing. *Journal of Applied Psychology*, 90 (3), 574–585.

Allen, N.J. and Meyer, J.P., 1996. Affective, continuance, and normative commitment to the organization: an examination of construct validity. *Journal of Vocational Behavior*, 49 (3), 252–276.

Ames, M. and Naaman, M., 2007. Why we tag: motivations for annotation in mobile and online media. *In*: *Proceedings of Computer/Human Interaction 2007*. 28 April – 3 May 2007, San Jose, California, USA.

Anderson, J.C. and Gerbing, D.W., 1988. Structural equation modeling in practice: A review and recommended two-step approach. *Psychological Bulletin*, 103 (3), 41–423.

Ashforth, B.E. and Mael, F., 1989. Social identity theory and the organization. *Academy of Management Review*, 14 (1), 20–39.

Bagozzi, R.B. and Dholakia, U.M., 2002. Intentional social action in virtual communities. *Journal of Interactive Marketing*, 16 (2), 2–79.

Bagozzi, R.B. and Dholakia, U.M., 2006. Open source software user communities: a study of participation in linux user groups. *Management Science*, 52 (7), 1099–1115.

Bagozzi, R.P. and Phillips, L.W., 1982. Representing and testing organizational theories: a holistic construal. *Administrative Science Quarterly*, 27 (3), 459–489.

Bargh, J.A., 1997. The automaticity of everyday life. *In*: S. Rober and J. Wyereds, eds. *Advances in social cognition*. Mahwah, New Jersey: Lawrence Erlbaum Associates, Vol. X, 1–62.

Baron, R.M. and Kenny, D.A., 1986. The moderator-mediator variable distinction in social psychological research: conceptual, strategic, and statistical consideration. *Journal of Personality and Social Psychology*, 51 (6), 1173–1182.

Baumeister, R.F., 2002. Yielding to temptation: Self-control failure, impulsive purchasing, and consumer behavior. *Journal of Consumer Research*, 28 (4), 670–676.

Bergami, M. and Bagozzi, R.P., 2000. Self-categorization, affective commitment and group self-esteem as distinct aspects of social identity in the organization. *British Journal of Social Psychology*, 39 (4), 555–577.

Biocca, F., Harms, C., and Burgoon, J.K., 2003. Toward a more robust theory and measure of social presence: review and suggested criteria. *Presence*, 12 (5), 456–480.

Biocca, F., Harms, C., and Gregg, J., 2001. The networked minds measure of social presence: pilot test of the factor structure and concurrent validity. *In*: *Presence 2001*, 21–23 May 2001, Philadelphia.

Biocca, F., Kim, T., and Levy, M.R., 1995. The vision of virtual reality. *In*: F. Biocca and M.R. Levy, eds. *Communication in the age of virtual reality*. Hillsdale: Lawrence Erlbaum Associates Inc, 3–14.

Browne, M.W. and Cudeck, R., 1993. Alternative ways of assessing model fit. *In*: K.A. Bollen and J.S. Long, eds. *Testing structural equation models*. Newbury Part, CA: Sage, 445–455.

Cameron, J., 2004. A three-factor model of social identity. *Self and Identity*, 3 (3), 239–262.

Chiu, C.M., Hsu, M.H., and Wang, E.T.G., 2006. Understanding knowledge sharing in virtual communities: An integration of social capital and social cognitive theories. *Decision Support Systems*, 42 (3), 1872–1888.

Christophel, D.M., 1990. The relationships among teacher immediacy behaviors, student motivation, and learning. *Communication Education*, 37, 323–340.

Chui, P.Y., Cheung, C.M.K., and Lee, M.K.O., 2008. Online social networks: Why do "we" use facebook? *In*: *First World Summit on the Knowledge Society (WSKS 2008)*, Athens, Greece.

Constant, D., Kiesler, S., and Sproull, L., 1994. What's mine is ours or is it? A study of attitudes about information sharing. *Information Systems Research*, 5 (4), 400–421.

Daft, R.L. and Lengel, R.H., 1986. Organizational information requirements, media richness and structural design. *Management Science*, 32 (5), 554–571.

Dholakia, U.M., Bagozzi, R.P., and Pearo, L.K., 2004. A social influence model of consumer participation in network- and small-group-based virtual communities. *International Journal of Research in Marketing*, 21 (3), 241–263.

Dukerich, J.M., Golden, B.R., and Shortell, S.M., 2002. Beauty is in the eye of the beholder: the impact of organizational identification, identity, and image on the cooperative behaviors of physicians. *Administrative Science Quarterly*, 47 (September), 507–533.

Ellemers, N., Kortekaas, P., and Ouwerkerk, J.W., 1999. Self-categorisation, commitment to the group and group self-esteem as related but distinct aspects of social identity. *European Journal of Social Psychology*, 29 (2–3), 371–389.

Flynn, F.J., 2005. Identity orientations and forms of social exchange in organizations. *Academy of Management Review*, 30 (4), 737–750.

Forehand, M.R., Deshpande, R., and Reed, A. II, 2002. Identity salience and the influence of differential activation of the social self-schema on advertising response. *Journal of Applied Psychology*, 87 (6), 1086–1099.

Foreman, P. and Whetten, D.A., 2002. Members' identification with multiple-identity organizations. *Organization Science*, 13 (6), 618–635.

Fornell, C. and Larcker, D., 1981. Evaluating structural equation models with unobservable variables and measurement error. *Journal of Marketing Research*, 18, 39–50.

Hair, J.F., *et al*., 1998. *Multivariate data analysis*, 5th ed. Upper Saddle River, NJ: Prentice Hall.

Haslam, S.A., *et al*., 1999. Social identity salience and the emergence of stereotype consensus. *Personality and Social Psychology Bulletin*, 25 (7), 809–818.

Hatcher, L., 1994. *A step-by-step approach to using the sas system for factor analysis and structural equation modeling*. Cary, NC: SAS Institute Inc.

Heeter, C., 1992. Being there: The subjective experience of presence. *Presence*, 1 (2), 262–271.

Heeter, C., 2003. Reflections on real presence by a virtual person. *Presence*, 12 (4), 335–345.

Hogg, M.A. and Abrams, D., 1988. *Social identifications: a social psychology of intergroup relations and group processes*. London: Routledge.

Huang, M.P., Norman, E., and Alessi, N.E., 1999. Presence as an emotional experience. *In*: J.D. Westwood, H.M. Hoffman, and R.A. Robb, eds. *Medicine meets virtual reality: the convergence of physical and informational technologies options for a new era in healthcare*. Amsterdam, The Netherlands: IOS Press, 148–153.

Ijsselsteijn, W.A. and Riva, G., 2003. Being there: the experience of presence in mediated environments. *In*: G. Riva, F. Davide, and W.A. Ijsselsteijn, eds. *Being there: concepts, effects and measurement of user presence in synthetic environments*. Amsterdam, The Netherlands: Ios Press, 3–16.

Jones, Q., Ravid, G., and Rafaeli, S., 2004. Information overload and the message dynamics of online interaction spaces: a theoretical model and empirical exploration. *Information Systems Research*, 15 (2), 194–210.

Kankanhalli, A., Tan, B.C.Y., and Wei, K.-K., 2005. Contributing knowledge to electronic knowledge repositories: an empirical investigation. *MIS Quarterly*, 29 (1), 113–143.

Kumar, N. and Benbasat, I., 2002. Para-social presence: a reconceptualization of 'social presence' to capture the relationship between a web site and her visitors. *In*: *Proceedings of the 35th Hawaii International Conference on System Sciences*, Hawaii, USA.

Lee, F.S.L., Vogel, D., and Limayem, M., 2003. Virtual community informatics: a review and research agenda. *Journal of Information Technology and Application*, 5 (1), 47–61.

Lee, K., 2006. What goes around comes around: An analysis of del.Icio.Us as social space. *In*: *20th Conference on Computer Supported Cooperative Work*, Banff, Alberta, Canada.

Leimeister, J.M., Sidiras, P., and Kremar, H.C., 2004. Success factors of virtual communities from the perspective of members and operators: an empirical study. *In*: *The 37th Hawaii International Conference on System Sciences*, Hawaii.

Lombard, M. and Ditton, T., 1997. At the heart of it all: The concept of presence. *Journal of Computer Mediated Communication*, 3 (2). Available from: http://www.ascusc.org/jcmc/vol3/issue2/lombard.html [Accessed 16 July 2009].

Ma, M. and Agarwal, R., 2007. Through a glass darkly: Information technology design, identity verification, and knowledge contribution in online communities. *Information Systems Research*, 18 (1), 42–67.

Mael, F. and Ashforth, B., 1992. Alumni and their alma mater: a partial test of the reformulated model of organizational identification. *Journal of Organizational Behavior*, 13, 103–123.

Mael, F. and Ashforth, B.E., 1992. Alumni and their alma mater: a partial test of the reformulated model of organizational identification. *Journal of Organizational Behavior*, 13 (2), 103–123.

Miranda, S.M. and Saunders, C.S., 2003. The social construction of meaning: an alternative perspective on information sharing. *Information Systems Research*, 14 (1), 87–106.

Nahapiet, J. and Ghoshal, S., 1998. Social capital, intellectual capital, and organizational advantage. *Academy of Management Review*, 23 (2), 242–266.

Nov, O., Naaman, M., and Ye, C., 2008. What drives content tagging: the case of photos on flickr. *In*: *ACM SIGCHI Conference on Human Factors in Computing Systems*, Florence, Italy.

Oakes, P.J., 1987. The salience of social categories. *In*: J.C. Turner, M.A. Hogg, P.J. Oakes, S.D. Reicher, and M.S. Wetherell, eds. *Rediscovering the social group: a self-categorization theory*. Oxford: Blackwell, 117–141.

Oakes, P.J., Haslam, S.A., and Turner, J.C., 1994. *Stereotyping and social reality*. Oxford, UK: Blackwell.

Pares, N. and Pares, R., 2006. Towards a model for a virtual reality experience: the virtual subjectiveness. *Presence*, 15 (5), 524–538.

Podsakoff, P.M., 1986. Self-reports in organizational research: problems and prospects. [No]. *Journal of Management*, 12 (4), 531–544.

Postmes, T., 2003. A social identity approach to communication in organizations. *In*: S.A. Haslam, D. van Knippenberg, and M.J. Platow, eds. *Social identity at work: developing theory for organizational practice*. Philadelphia, PA: Psychology Press, 81–97.

Prasolova-Forland, E., 2002. Supporting social awareness in education in collaborative virtual environments. *In*: *CSCW'02*, , 16–20 November 2002.

Puddifoot, J.E., 2003. Exploring personal and shared sense of community identity in Durham city, England. *Journal of Community Psychology*, 31 (1), 81–106.

Reis, H.T. and Shaver, P., 1988. Intimacy as an interpersonal process. *In*: S. Duck, ed. *Handbook of personal relationship*. Chichester, England: Wiley, 367–389.

Richmond, V.P., 1990. Communication in the classroom: power and motivation. *Communication Education*, 39, 181–195.

Riva, G., *et al*., 2002. Towards a cultural approach to presence. *In*: *Presence 2002*, Porto, Portugal.

Ryan, R.M. and Deci, E.L., 2000. Self-determination theory and the facilitation of intrinsic motivation, social development, and well-being. *American Psychologist*, 55 (1), 68–78.

Ryan, R.M. and Deci, E.L., 2001. To be happy or to be self-fulfilled: a review of research on hedonic and eudaimonic well-being. *Annual Review of Psychology*, 52, 141–166.

Shen, K.N. and Khalifa, M., 2007. Exploring multi-dimensional conceptualization of social presence in the context of online communities. *In*: *Human-Computer Interaction International 2007*, Beijing, China.

Short, J., Williams, E., and Christie, B., 1976. *The social psychology of telecommunications*. London: Wiley.

Smith, E.R. and Henry, S., 1996. An in-group becomes part of the self: response time evidence. *Personality and Social Psychology Bulletin*, 22, 635–642.

Teo, T.S.H. and King, W.R., 1996. Key dimensions of facilitators and inhibitors for the strategic use of information technology. *Journal of Management Information Systems*, 12 (4), 35–53.

Turner, J.C., 1982. Towards a cognitive redefinition of the social group. *In*: H. Tajfel, ed. *Social identity and intergroup relations*. Cambridge: Cambridge University Press, 15–40.

Turner, J.C., *et al*., 1994. Self and collective: cognition and social context. *Personality and Social Psychology Bulletin*, 20 (5), 454–463.

Van Der Vegt, G.S. and Bunderson, J.S., 2005. Learning and performance in multidisciplinary teams: the performance of collective team identification. *Academy of Management Journal*, 48 (3), 532–547.

van Knippenberg, D., 2000. Work motivation and performance: a social identity perspective. *Applied Psychology: An International Review*, 49 (3), 357–371.

Vastfjall, D., 2003. The subjective sense of presence, emotion recognition, and experienced emotions in auditory virtual environment. *CyberPsychology & Behavior*, 6 (2), 181–188.

Wasko, M. and Faraj, S., 2000. It is what one does: Why people participate and help others in electronic communities of practice. *Journal of Strategic Information Systems*, 9 (2–3), 155–173.

Wasko, M.M. and Faraj, S., 2005. Why should I share? Examining social capital and knowledge contribution in electronic networks of practice. *MIS Quarterly*, 29 (1), 35–57.

Weber, J.M., Kopelman, S., and Messick, D.M., 2004. A conceptual review of decision making in social dilemmas: applying a logic of appropriateness. *Personality and Social Psychology Review*, 8 (3), 281–307.

Wenger, E. and Snyder, W., 2000. Communities of practice: the organizational frontier. *Harvard Business Review*, 78 (1), 139–145.